

Lecture 1: Hartree–Fock Theory

1. The Electronic Structure Problem

Everything in the next five lectures follows from one equation. A molecule of N electrons and L nuclei is described, in atomic units, by

$$\hat{H} = -\sum_i \frac{1}{2} \nabla_i^2 - \sum_I \frac{1}{2M_I} \nabla_I^2 - \sum_{i,I} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \sum_{i<j} \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} + \sum_{I<J} \frac{Z_I Z_J}{|\mathbf{R}_I - \mathbf{R}_J|}. \quad (1)$$

Atomic units set $\hbar = m_e = e = 4\pi\epsilon_0 = 1$, which is why the kinetic operator is $-\frac{1}{2}\nabla^2$ and every Coulomb interaction is a bare $1/r$. Lengths are in bohr ($a_0 \approx 0.529 \text{ \AA}$), energies in hartree ($E_h \approx 27.21 \text{ eV} \approx 627.5 \text{ kcal/mol}$), and nuclear masses in electron masses, so $M_I \gtrsim 1836$. Note that chemical accuracy, $1 \text{ kcal/mol} \approx 1.6 \text{ m}E_h$, is about a part in 10^5 of the total energy of a small molecule, which is why we care about energy *differences* far more than total energies.

Because nuclei are thousands of times heavier than electrons, we invoke the Born–Oppenheimer approximation and factorize

$$\Psi(\{\mathbf{r}_i\}, \{\mathbf{R}_I\}) = \psi_{\{\mathbf{R}_I\}}(\{\mathbf{r}_i\}) \chi(\{\mathbf{R}_I\}), \quad (2)$$

solving the *electronic* problem at fixed nuclear positions,

$$\hat{H}_{\text{el}}(\{\mathbf{R}_I\}) \psi_k = E_k(\{\mathbf{R}_I\}) \psi_k, \quad \hat{H}_{\text{el}} = -\frac{1}{2} \sum_i \nabla_i^2 - \sum_{i,I} \frac{Z_I}{|\mathbf{r}_i - \mathbf{R}_I|} + \sum_{i<j} \frac{1}{r_{ij}} + \sum_{I<J} \frac{Z_I Z_J}{R_{IJ}}. \quad (3)$$

The nuclear coordinates enter only as parameters. The function $E_0(\{\mathbf{R}_I\})$ is the ground-state potential energy surface, the object that molecular dynamics propagates on. Our entire job in this course is to compute $E_k(\{\mathbf{R}_I\})$ as accurately and as cheaply as possible.

2. Why This Is Hard

Equation (3) is a partial differential equation in $3N$ dimensions. If we discretize each spatial dimension on N_g grid points, the wavefunction $\psi(\mathbf{r}_1, \dots, \mathbf{r}_N)$ requires

$$\mathcal{O}(N_g^{3N}) \quad (4)$$

numbers to store. For a water molecule ($N = 10$) with a modest grid this already exceeds the number of atoms in the universe. This is the *curse of dimensionality*, and it is the reason quantum chemistry exists as a field: we must find compressed, physically motivated parameterizations of ψ .

Dirac put it best in 1929: the underlying laws are completely known, “and the difficulty is only that the exact application of these laws leads to equations much too complicated to be soluble.”

3. Antisymmetry and Slater Determinants

The first physical constraint that compresses the problem is the Pauli principle. Electrons are spin-1/2 fermions, so with $\tau \equiv (\mathbf{r}, \sigma)$ the permutation operator satisfies

$$\hat{P}_{12} \Psi(\tau_1, \tau_2) = \Psi(\tau_2, \tau_1) = -\Psi(\tau_1, \tau_2). \quad (5)$$

Since $[\hat{H}_{\text{el}}, \hat{P}_{ij}] = 0$, exact eigenstates may be chosen as eigenstates of $\hat{P}_{ij}$, and for electrons the eigenvalue must be -1 . We require the *same* of any approximate wavefunction.

The simplest antisymmetric ansatz built from one-electron spin orbitals $\phi_p(\tau) = \psi_p(\mathbf{r}) \omega_p(\sigma)$ is the Slater determinant

$$\Psi_{\text{SD}}(\tau_1, \dots, \tau_N) = \frac{1}{\sqrt{N!}} \begin{vmatrix} \phi_1(\tau_1) & \cdots & \phi_N(\tau_1) \\ \vdots & \ddots & \vdots \\ \phi_1(\tau_N) & \cdots & \phi_N(\tau_N) \end{vmatrix} = \sqrt{N!} \hat{A} [\phi_1 \phi_2 \cdots \phi_N], \quad (6)$$

where the antisymmetrizer is

$$\hat{A} = \frac{1}{N!} \sum_{K \in S_N} (-1)^{P_K} \hat{P}_K, \quad \hat{A}^2 = \hat{A}. \quad (7)$$

The $\sqrt{N!}$ in front of $\hat{A}$ is bookkeeping: expanding the determinant produces $\frac{1}{\sqrt{N!}} \sum_K (-1)^{P_K} \hat{P}_K$, and with $\hat{A}$ normalized as an idempotent projector the two factors of $N!$ combine to leave $\sqrt{N!}$. Interchanging two rows flips the sign, and two identical columns give zero: the Pauli principle is built in. Note that a determinant is *not* a product state: it carries the minimal “exchange” correlation forced by antisymmetry, but no more. Everything missing beyond it we will call electron correlation.

4. Slater–Condon Rules

To use determinants we need matrix elements. Writing $\hat{H}_{\text{el}} = \sum_i \hat{h}(i) + \sum_{i < j} r_{ij}^{-1} + E_{\text{nuc}}$ with orthonormal spin orbitals, the rules are:

$$\langle \Psi | \hat{H} | \Psi \rangle = \sum_i^{\text{occ}} h_{ii} + \frac{1}{2} \sum_{ij}^{\text{occ}} \langle ij || ij \rangle + E_{\text{nuc}}, \quad (8)$$

$$\langle \Psi | \hat{H} | \Psi_i^a \rangle = h_{ia} + \sum_j^{\text{occ}} \langle ij || aj \rangle \equiv F_{ia}, \quad (9)$$

$$\langle \Psi | \hat{H} | \Psi_{ij}^{ab} \rangle = \langle ij || ab \rangle, \quad (10)$$

and matrix elements between determinants differing by three or more spin orbitals vanish, because $\hat{H}$ contains at most two-body operators. Here $|\Psi_i^a\rangle$ denotes the determinant with ϕ_i replaced by ϕ_a , and

$$\langle pq || rs \rangle \equiv \langle pq | rs \rangle - \langle pq | sr \rangle, \quad \langle pq | rs \rangle = \int d\tau_1 d\tau_2 \phi_p^*(\tau_1) \phi_q^*(\tau_2) \frac{1}{r_{12}} \phi_r(\tau_1) \phi_s(\tau_2). \quad (11)$$

The chemists’ notation for the same integral is $(pr|qs) = \langle pq | rs \rangle$.

A two-electron example. For a closed-shell two-electron system with both electrons in the spatial orbital ψ_1 , the occupied spin orbitals are $\psi_1\alpha$ and $\psi_1\beta$. Equation (8) gives

$$E_{\text{el}} = 2h_{11} + (11|11), \quad (12)$$

because the direct term appears for both $\alpha\alpha$, $\beta\beta$, $\alpha\beta$ and $\beta\alpha$ pairs, while the exchange integral between opposite spins carries the factor $\langle \alpha | \beta \rangle = 0$ and vanishes. Exchange acts only between electrons of like spin.

5. Linear Combination of Atomic Orbitals

We still need the orbitals. In practice we expand them in a fixed set of M basis functions,

$$\psi_i(\mathbf{r}) = \sum_{\mu=1}^M C_{\mu i} \chi_{\mu}(\mathbf{r}), \quad (13)$$

which turns a differential equation into a matrix problem. Common choices:

- **Atom-centered functions.** Slater-type ($e^{-\alpha r}$), Gaussian-type ($e^{-\alpha r^2}$), or numerical radial parts times Y_{lm} . Gaussians dominate molecular codes because all required integrals are analytic.
- **Plane waves** $e^{i\mathbf{k}\cdot\mathbf{r}}$, standard in solid state.
- **Grids and finite elements**, systematically improvable but expensive.

Why the mixing in (13) is essential: for H_2 in a minimal basis, the determinant built from the two *atomic* $1s$ functions gives $E = -0.8807$ a.u. at $R = 0.741 \text{ \AA}$, which lies above the separated-atom limit of -1 a.u., i.e. no bond at all. Forming the symmetry-adapted molecular orbitals

$$\psi_g \propto \chi_{1s}^A + \chi_{1s}^B, \quad \psi_u \propto \chi_{1s}^A - \chi_{1s}^B, \quad (14)$$

and doubly occupying ψ_g yields $E = -1.1336$ a.u., which does bind. The question is how to find the coefficients $C_{\mu i}$ in general.

6. The Variational Principle and the Roothaan–Hall Equations

The Hartree–Fock energy is the best single-determinant energy,

$$E_{\text{HF}} = \min_{\mathbf{C}} \frac{\langle \Psi_{\text{SD}} | \hat{H} | \Psi_{\text{SD}} \rangle}{\langle \Psi_{\text{SD}} | \Psi_{\text{SD}} \rangle}. \quad (15)$$

Introduce the one-particle density matrix $P_{\mu\nu} = \sum_i^{\text{occ}} C_{\mu i} C_{\nu i}^*$, i.e. $\mathbf{P} = \mathbf{C}_{\text{occ}} \mathbf{C}_{\text{occ}}^\dagger$. Then

$$E = \sum_{\mu\nu} P_{\nu\mu} h_{\mu\nu} + \frac{1}{2} \sum_{\mu\nu\lambda\sigma} P_{\nu\mu} P_{\sigma\lambda} [(\mu\nu|\lambda\sigma) - (\mu\sigma|\lambda\nu)] + E_{\text{nuc}}. \quad (16)$$

Minimizing under the orthonormality constraint $\langle i|j \rangle = \sum_{\mu\nu} C_{\mu i}^* S_{\mu\nu} C_{\nu j} = \delta_{ij}$ requires the Lagrangian

$$\mathcal{L} = E - \sum_{ij} \Lambda_{ij} (\langle i|j \rangle - \delta_{ij}). \quad (17)$$

Setting $\partial\mathcal{L}/\partial C_{\mu i}^* = 0$ gives $\mathbf{FC} = \mathbf{SCA}$, where the Fock matrix is the derivative of the energy with respect to the density,

$$F_{\mu\nu} = \frac{\partial E}{\partial P_{\nu\mu}} = h_{\mu\nu} + \underbrace{\sum_{\lambda\sigma} P_{\sigma\lambda} (\mu\nu|\lambda\sigma)}_{J_{\mu\nu}} - \underbrace{\sum_{\lambda\sigma} P_{\sigma\lambda} (\mu\sigma|\lambda\nu)}_{K_{\mu\nu}}. \quad (18)$$

Because $\mathbf{A}$ is Hermitian we may diagonalize it, $\mathbf{U}^\dagger \mathbf{A} \mathbf{U} = \boldsymbol{\varepsilon}$; the energy is invariant under this rotation among occupied orbitals, so we are free to work in the *canonical* basis $\tilde{\mathbf{C}} = \mathbf{C} \mathbf{U}$:

$$\boxed{\mathbf{F} \mathbf{C} = \mathbf{S} \mathbf{C} \boldsymbol{\varepsilon}} \quad (19)$$

This is the Roothaan–Hall equation (1951). Note the essential nonlinearity: $\mathbf{F} = \mathbf{F}[\mathbf{P}[\mathbf{C}]]$. Each electron moves in the mean field of all the others, which is why Hartree–Fock is called a mean-field theory.

7. Solving the SCF Equations

Equation (19) is a generalized eigenvalue problem because the basis is not orthogonal. We remove $\mathbf{S}$ with a transformation $\mathbf{X}$ satisfying $\mathbf{X}^\dagger \mathbf{S} \mathbf{X} = \mathbf{I}$; the two standard choices are Löwdin’s $\mathbf{X} = \mathbf{S}^{-1/2}$ and canonical orthogonalization $\mathbf{X} = \mathbf{U} \boldsymbol{\sigma}^{-1/2}$ from $\mathbf{S} = \mathbf{U} \boldsymbol{\sigma} \mathbf{U}^\dagger$, where near-zero eigenvalues of $\mathbf{S}$ are discarded (linear dependence). Then

$$\mathbf{F}' \mathbf{C}' = \mathbf{C}' \boldsymbol{\varepsilon}, \quad \mathbf{F}' = \mathbf{X}^\dagger \mathbf{F} \mathbf{X}, \quad \mathbf{C} = \mathbf{X} \mathbf{C}'. \quad (20)$$

The self-consistent field (SCF) loop is:

1. Guess $\mathbf{C}$, form $\mathbf{P} = \mathbf{C}_{\text{occ}} \mathbf{C}_{\text{occ}}^\dagger$.
2. Build $\mathbf{F}[\mathbf{P}]$.
3. Diagonalize to get new orbitals; occupy the N lowest (Aufbau).
4. Stop when $|\Delta E| < \varepsilon_1$ ($\sim 10^{-6}$) and $\|\Delta \mathbf{P}\| < \varepsilon_2$ ($\sim 10^{-4}$); otherwise return to step 2.

Plain fixed-point iteration often oscillates. The standard accelerator is DIIS (Pulay, 1980), which extrapolates the Fock matrix so as to minimize the norm of the residual

$$\mathbf{R} = \mathbf{F} \mathbf{P} \mathbf{S} - \mathbf{S} \mathbf{P} \mathbf{F}. \quad (21)$$

This is the right error metric because it vanishes exactly at a solution: from $\mathbf{F} \mathbf{C} = \mathbf{S} \mathbf{C} \boldsymbol{\varepsilon}$ we get $\mathbf{F} \mathbf{P} \mathbf{S} = \mathbf{F} \mathbf{C}_{\text{occ}} \mathbf{C}_{\text{occ}}^\dagger \mathbf{S} = \mathbf{S} \mathbf{C}_{\text{occ}} \boldsymbol{\varepsilon}_{\text{occ}} \mathbf{C}_{\text{occ}}^\dagger \mathbf{S}$, which is manifestly Hermitian, so it equals its own adjoint $\mathbf{S} \mathbf{P} \mathbf{F}$. Alternatives include Broyden mixing and direct minimization on the orbital manifold (GDM).

8. What the Orbitals Mean

The orbital energies are not observables, but they are close to some. From $\boldsymbol{\varepsilon} = \mathbf{C}^\dagger \mathbf{F} \mathbf{C}$,

$$\varepsilon_i = h_{ii} + \sum_j^{\text{occ}} \langle ij || ij \rangle \implies \sum_i^{\text{occ}} \varepsilon_i = \sum_i h_{ii} + \sum_{ij} \langle ij || ij \rangle \neq E_{\text{HF}}, \quad (22)$$

the two-electron terms being double counted in the orbital-energy sum.

Koopmans' theorem (1934). Remove the electron in orbital k and *freeze* the remaining orbitals. Then

$$E_{\text{HF}}(N) - E_{\text{HF}}(N-1) = h_{kk} + \sum_j \langle kj || kj \rangle = \varepsilon_k, \quad (23)$$

so that

$$\text{IP} = E(N-1) - E(N) \approx -\varepsilon_{\text{HOMO}}, \quad \text{EA} = E(N) - E(N+1) \approx -\varepsilon_{\text{LUMO}}, \quad (24)$$

and the fundamental (quasiparticle) gap is $\text{IP} - \text{EA} \approx \varepsilon_{\text{LUMO}} - \varepsilon_{\text{HOMO}}$. The approximation neglects orbital relaxation of the ion (which lowers the true IP) and correlation (which raises it); the two errors partially cancel, which is why Koopmans works as well as it does. ΔSCF , i.e. separate SCF calculations on the N - and $(N-1)$ -electron states, restores relaxation.

Brillouin's theorem. Consider rotating an occupied orbital i into a virtual orbital a by an angle κ . For a closed-shell reference the energy gradient is

$$\left. \frac{dE}{d\kappa_{ia}} \right|_{\kappa=0} = 4F_{ia}. \quad (25)$$

At a stationary point the gradient vanishes, so $F_{ia} = 0$ for all i, a , and by the Slater–Condon rule (9),

$$\langle \Psi_i^a | \hat{H} | \Psi_{\text{HF}} \rangle = F_{ai} = 0. \quad (26)$$

Singly excited determinants do not couple to a converged HF reference. This one fact will simplify everything we do in Lectures 3 and 4.

9. A Two-Orbital Example (optional)

Almost every qualitative feature of SCF theory survives in the symmetric two-basis function problem. Take

$$\mathbf{F} = \begin{pmatrix} f & g \\ g & f \end{pmatrix}, \quad \mathbf{S} = \begin{pmatrix} 1 & s \\ s & 1 \end{pmatrix}, \quad |s| < 1. \quad (27)$$

Symmetry dictates the eigenvectors, and $\mathbf{S}$ -normalization ($\mathbf{c}^\dagger \mathbf{S} \mathbf{c} = 1$) fixes their length:

$$\mathbf{c}_\pm = \frac{1}{\sqrt{2(1 \pm s)}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}, \quad \varepsilon_\pm = \frac{f \pm g}{1 \pm s}. \quad (28)$$

For $g < 0$ (a typical resonance integral) $\mathbf{c}_+$ is the bonding orbital. Note the asymmetry: the antibonding level is pushed up more than the bonding level is pushed down, entirely because of the overlap s . This is the origin of the fact that four-electron interactions are repulsive.

10. RHF, UHF, GHF, and the Symmetry Dilemma (optional)

We have been sloppy about spin. Writing the MO coefficients and density in spin blocks,

$$\mathbf{P} = \begin{pmatrix} \mathbf{P}_{\alpha\alpha} & \mathbf{P}_{\alpha\beta} \\ \mathbf{P}_{\beta\alpha} & \mathbf{P}_{\beta\beta} \end{pmatrix}, \quad (29)$$

Fukutome's classification (1974) gives a hierarchy of constraints:

- **RHF**: $\mathbf{P}_{\alpha\alpha} = \mathbf{P}_{\beta\beta}$, $\mathbf{P}_{\alpha\beta} = 0$. Eigenfunction of $\hat{S}^2$ and $\hat{S}_z$.
- **UHF**: $\mathbf{P}_{\alpha\alpha} \neq \mathbf{P}_{\beta\beta}$, $\mathbf{P}_{\alpha\beta} = 0$. Eigenfunction of $\hat{S}_z$ only.
- **GHF**: no constraint; neither $\hat{S}^2$ nor $\hat{S}_z$ is good.

Since the constrained solutions are contained in the unconstrained ones, $E_{\text{GHF}} \leq E_{\text{UHF}} \leq E_{\text{RHF}}$. This creates Löwdin’s *symmetry dilemma*: the exact wavefunction is an eigenfunction of every symmetry that commutes with $\hat{H}$, but we can always lower the mean-field energy by breaking one.

Spin contamination, quantitatively. Take a two-electron UHF determinant with

$$\psi_\alpha(\theta) = \cos \theta \psi_g + \sin \theta \psi_u, \quad \psi_\beta(\theta) = \cos \theta \psi_g - \sin \theta \psi_u, \quad 0 \leq \theta \leq \pi/4. \quad (30)$$

Then $\langle \psi_\alpha | \psi_\beta \rangle = \cos^2 \theta - \sin^2 \theta = \cos 2\theta$, and using $\langle \hat{S}^2 \rangle = 1 - |\langle \psi_\alpha | \psi_\beta \rangle|^2$ for this two-electron case,

$$\langle \hat{S}^2 \rangle = 1 - \cos^2 2\theta = \sin^2 2\theta. \quad (31)$$

At $\theta = 0$ we recover the pure singlet RHF determinant ($\langle \hat{S}^2 \rangle = 0$); at $\theta = \pi/4$ the two electrons localize on opposite atoms and $\langle \hat{S}^2 \rangle = 1$, a 50/50 singlet–triplet mixture. This is exactly what happens when H_2 is stretched: beyond the Coulson–Fischer point the RHF solution becomes unstable, UHF breaks spin symmetry and gives the correct energy at dissociation, but the wavefunction is no longer a spin eigenstate. Restricted HF, by contrast, dissociates H_2 to a 50/50 mixture of the covalent and ionic ($\text{H}^+ \cdots \text{H}^-$) configurations and is catastrophically too high in energy.

11. Cost and Accuracy

There are formally $\mathcal{O}(M^4)$ two-electron integrals, but with distance and Schwarz screening only $\mathcal{O}(M^2)$ are numerically significant, so the asymptotic bottleneck is the $\mathcal{O}(M^3)$ diagonalization. On a single node, $\sim 10,000$ basis functions and $\sim 1,000$ electrons are routine.

What do we get for it? Hartree–Fock recovers about 99% of the total electronic energy, and is still useless for chemistry:

Table 1: Mean absolute errors (kcal/mol). Atomization energies are for the G2 set of Curtiss et al. (>50 molecules), as quoted by Koch and Holthausen; barrier heights are from the HTBH38/04 database (38 hydrogen transfer barriers) and from the heavy-atom transfer subset of NHTBH38/04 (12 barriers), Zhao, González-García and Truhlar.

Method	G2 atomization	Barrier heights	
	energies	H transfer	Heavy-atom transfer
HF	78	13.7	16.9
B3LYP	~ 2	4.2	8.5

The missing 1% is chemistry. We define the *correlation energy* as

$$E_{\text{corr}} = E_{\text{exact}} - E_{\text{HF}}, \quad (32)$$

where “exact” means the exact solution of (3) in the same basis (full configuration interaction). It is always negative, since HF is variational. Chemical accuracy is 1 kcal/mol $\approx$ 1.6 mHartree $\approx$ 0.04 eV; bond energies are of order 100 kcal/mol. We are chasing errors at the 10^{-3} level of the total energy.

12. Where We Go From Here

Hartree–Fock is the reference point for everything that follows. Two roads lead out of it:

- Keep the mean-field structure and the $\mathcal{O}(N^3)$ cost, but replace the exchange interaction by an approximate functional of the density that also folds in correlation. That is density functional theory (Lecture 2).
- Keep the Hamiltonian exact and systematically expand the wavefunction beyond a single determinant. That is perturbation theory, configuration interaction, and coupled cluster (Lecture 3), and their excited-state analogues (Lecture 4).

Existing quantum chemistry approaches struggle with strong correlation, which is where quantum computation enters the story (Lecture 5).